

PREDICTIVE MODELING AND RISK SCORING FOR BANK CUSTOMER CHURN

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Abstract—Customer churn is one of the major challenges faced by banks, as losing existing customers can significantly impact profitability and long-term business growth. The objective of this project is to develop a predictive modeling and risk scoring system that identifies customers who are likely to leave the bank using historical customer data. The project begins with data preprocessing and Exploratory Data Analysis (EDA) to understand customer behavior and identify key factors associated with churn. Three random forests, Random Forest, were developed and evaluated to predict customer churn. The performance of each model was compared using standard evaluation metrics, and the Random Forest model achieved the highest prediction accuracy of 86.6%, making it the most suitable model for this study. The developed predictive system enables banks to identify high-risk customers at an early stage and supports data-driven retention strategies, helping improve customer satisfaction and reduce customer attrition.

Keywords—customer churn, predictive modeling, risk scoring, machine learning, banking analytics, random forest, customer retention, data analytics.

I. INTRODUCTION

Customer churn has become one of the most significant challenges in the banking industry, as retaining existing customers is often more cost-effective than acquiring new ones. When customers discontinue their relationship with a bank, it not only results in financial losses but also affects customer loyalty, market competitiveness, and long-term business growth. Therefore, Identifying customers who are likely to leave the bank before they churn has become an important objective for financial institutions. With the increasing availability of customer data,

predictive analytics and machine learning techniques have emerged as effective solutions for understanding customer behavior and supporting data-driven decision-making.

This research presents a predictive modeling and risk scoring approach for bank customer churn using machine learning techniques. The project utilizes historical banking customer data to analyze customer characteristics, identify important factors associated with churn, and develop predictive models capable of estimating churn risk. Exploratory Data

Analysis (EDA) was performed to understand the dataset and uncover meaningful insights before model development. Three machine learning algorithms, namely logistic regression, random forest, and random forest, were implemented and evaluated to identify the most suitable model for customer churn prediction. The findings of this study aim to assist banks in developing proactive customer retention strategies and improving overall business performance through data-driven insights.

II. PROBLEM STATEMENT

Customer churn has become a major concern for banking institutions due to its direct impact on revenue, customer loyalty, and long-term business growth. Acquiring a new customer is significantly more expensive than retaining an existing one, making customer retention a critical business objective. However, banks often struggle to identify customers who are likely to discontinue their services before the actual churn occurs. As a result, valuable customers may be lost without timely intervention.

With the rapid growth of customer data, banks have the opportunity to leverage predictive analytics and machine learning to identify potential churners at an early stage. An accurate churn prediction system enables financial institutions to assess customer risk, implement model modeling and retention strategies, and optimize business decision-making. Therefore, there is a need for an intelligent predictive modelling and risk scoring system that can accurately estimate customer churn risk and support proactive customer retention efforts.

III. OBJECTIVES

The primary objective of this system is modeling and risk scoring that can accurately identify bank customers who are likely to churn based on historical customer data. Early prediction of customer churn enables banks to take proactive retention measures, improve customer satisfaction, and minimize revenue loss.

The specific objectives of this study are as follows:

- To analyze the banking customer dataset and identify the key factors influencing customer churn.
- To perform Exploratory Data Analysis (EDA) to understand customer behavior, detect patterns, and extract meaningful insights.
- To preprocess the dataset by encoding categorical variables and preparing the data for machine learning.
- To develop predictive models using logistic regression, decision tree, and random forest algorithms.
- To evaluate and compare the performance of the developed models using appropriate evaluation metrics.
- To estimate customer churn risk and provide business recommendations that support customer retention strategies and data-driven decision-making.

IV. DATASET DESCRIPTION

The study was conducted using the European Bank Customer Churn dataset, which contains historical information about bank customers and their account-related attributes. The dataset consists of 10,000 customer records and 14 variables representing behavioral behavioural characteristics that may influence customer churn.

The dataset includes features such as Credit Score, Geography, Gender, Age, Tenure, Account Balance, Number of Products, Credit Card Status, Active Membership, and Estimated Salary. The target variable, **Exited**, indicates whether a customer has left the bank, where a value of **1** represents a churned customer and **0** represents a retained customer.

Before model development, the dataset was examined to understand its structure and quality. The initial analysis confirmed that the dataset contained complete records without missing values, making it suitable for further preprocessing, exploratory data analysis, and predictive modeling. The diversity of customer attributes provides a strong foundation for identifying churn patterns and estimating customer risk using machine learning techniques.

V. METHODOLOGY

The methodology adopted in this study follows a systematic machine learning workflow for predicting customer churn and estimating customer risk in the banking sector. The research process consists of data collection, data preprocessing, exploratory data analysis, feature engineering, predictive model development, and model evaluation. Each stage was designed to ensure data quality, improve model performance, and generate meaningful business insights for effective customer retention strategies.

5.1 Data Collection

The study was conducted using the European Bank Customer Churn dataset, which contains historical customer information collected from a banking environment. The dataset includes 10,000 customer records with

demographic, financial, and account-related attributes that influence customer churn behavior. The dataset was selected because it provides a comprehensive representation of customer characteristics required for predictive modeling and risk analysis.

5.2 Data Preprocessing

Before developing the predictive models, the dataset was preprocessed to improve data quality and ensure compatibility with machine learning algorithms. The initial inspection confirmed that the dataset contained no missing values. Categorical variables such as geography and gender were encoded into numerical representations to make them suitable for model training. Irrelevant identifiers that do not contribute to prediction, such as customer identification fields, were excluded from the modeling performance of the predictive models on unseen data.

5.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset and identify patterns associated with customer churn. Statistical summaries and visualizations were used to examine the distribution of variables such as age, balance, geography, gender, tenure, and account activity. Correlation analysis and graphical representations helped identify relationships between customer attributes and the target variable, providing valuable insights before model development.

5.4 Feature Engineering

Feature engineering was performed to transform the dataset into a suitable format for machine learning algorithms. Categorical variables were encoded into numerical values, and irrelevant columns that did not contribute to customer churn prediction were excluded from the analysis. The dataset was then divided into training and testing subsets to ensure unbiased model evaluation. These

preprocessing steps improved the efficiency and predictive capability of the machine learning models.

5.5 Machine Learning Models

To predict customer churn, three supervised machine learning algorithms were implemented and compared in this study. Logistic regression was used as a baseline classification model due to its simplicity and interpretability. A decision tree patterns within the dataset. Finally, the Random Forest algorithm was implemented as an ensemble learning technique to improve predictive accuracy and reduce the risk of overfitting. The performance of all models was evaluated using the same training and testing datasets to ensure a fair comparison.

5.6 Model Evaluation

The developed machine learning models were evaluated using standard classification performance metrics. Model accuracy was used as the primary metric for comparing overall predictive performance. In addition, confusion matrices and classification reports were generated to assess the ability of each model to correctly classify churned and retained customers. The comparative evaluation enabled the selection of the most effective model for customer churn prediction and risk scoring.

VI. RESULTS AND DISCUSSION

The results obtained from Exploratory Data Analysis (EDA) and predictive modeling provide valuable insights into customer churn behavior. This section presents the key findings from the data analysis, compares the performance of different machine learning models, and discusses the business implications of the results.

6.1 Exploratory Data Analysis Findings

The Exploratory Data Analysis (EDA) revealed several important patterns associated with customer churn. The visual analysis indicated that customer churn is influenced by multiple demographic and financial factors rather than a single attribute. Variables such as age, account balance, geography, and active membership showed noticeable differences between customers who exited the bank and those who remained. Correlation analysis further highlighted the relationships among different variables, providing useful insights for predictive model development.

6.2 Model Performance Comparison

Three supervised machine learning algorithms were implemented and evaluated for predicting customer churn. Logistic Regression served as the baseline classification model and achieved an accuracy of 81.1%. The decision tree model obtained an accuracy of 77.45%, demonstrating its ability to capture non-linear relationships within the dataset. The Random Forest classifier produced the highest predictive performance with an accuracy of 86.6%, outperforming the other two models.

The superior performance of the Random Forest algorithm can be attributed to its ensemble learning approach, which combines multiple decision trees to improve prediction accuracy and reduce overfitting. Based on the comparative evaluation, the Random Forest model was selected as the final model for customer churn prediction and customer risk scoring.

Model	Accuracy
Logistic Regression	81.10%
Decision Tree	77.45%
Random Forest	86.60%

Model	Accuracy
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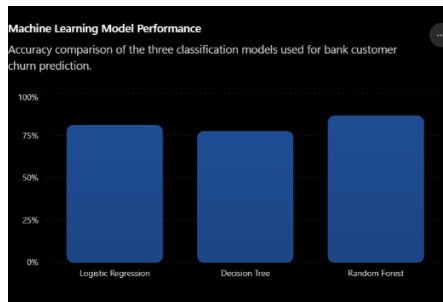


Figure 6 compares the predictive performance of the three machine learning algorithms implemented in this study. The Random Forest classifier achieved the highest accuracy of 86.6%, outperforming Logistic Regression (81.1%) and Decision Tree (77.45%). Based on this comparison, Random Forest was selected as the final model for customer churn prediction and risk scoring.

6.3 Feature Importance Analysis

To improve model interpretability, feature importance analysis was performed using the Random Forest classifier. The analysis identified the most influential variables contributing to customer churn prediction. Among all the features, age emerged as the most significant predictor with an importance score of 23.8%, followed by estimated salary, credit score, balance, and number of products. These variables play a crucial role in distinguishing customers who are likely to churn from those who remain with the bank.

The feature importance results provide valuable business insights by highlighting the customer characteristics that should be

monitored more closely. Banks can utilize these findings to design targeted retention strategies for customers exhibiting high-risk attributes.

6.4 Business Insights

The findings of this study indicate that customer churn is influenced by a combination of demographic, financial, and behavioral variations in churn across different customer segments, while feature importance analysis confirmed that Age, Estimated Salary, Credit Score, Balance, and Number of Products were among the strongest predictors of churn.

The comparative evaluation of machine learning models demonstrated that the Random Forest classifier achieved the best predictive performance with an accuracy of 86.6%. These results highlight the potential of machine learning in supporting proactive customer retention strategies. By identifying high-risk customers before they leave, banks can implement personalized retention campaigns, improve customer satisfaction, and reduce long-term revenue loss.

VII. BUSINESS RECOMMENDATIONS

Based on the findings of this study, the following recommendations are proposed for banking institutions:

- Utilize the Random Forest predictive model to identify customers with a high probability of churn.
- Develop personalized retention programs for high-risk customers through loyalty rewards, customized financial products, and proactive customer support.

- Monitor critical customer attributes such as age, credit score, balance, number of products, and

estimated salary

- Increase customer engagement by promoting active account usage and strengthening digital banking services.

- Integrate predictive analytics into the bank's Customer Relationship Management (CRM) system to support real-time churn monitoring and data-driven business decisions.

through personalized retention strategies. The study demonstrates the practical application of machine learning in banking analytics and highlights its value in supporting customer retention and business decision-making. Future work may focus on advanced ensemble techniques, real-time predictive systems, and deep learning approaches to further improve model performance.

VIII. CONCLUSION

This study developed a predictive modeling and risk scoring framework for identifying customers who are likely to churn in the banking sector. A comprehensive analysis was performed using the European Bank Customer Churn dataset, followed by data preprocessing, exploratory data analysis, feature engineering, and the implementation of three machine learning algorithms.

The experimental results demonstrated that the Random Forest classifier achieved the highest predictive accuracy of 86.6%, outperforming Logistic Regression (81.1%) and Decision Tree (77.45%). Feature importance analysis further revealed that age, estimated salary, credit score, balance, and number of products were the most influential factors affecting customer churn.

The proposed predictive framework enables banks to identify high-risk customers at an early stage, allowing timely intervention

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